

4(a)	<i>codon that</i> terminates / AW, translation / polypeptide synthesis R stops production of amino acids or does not specify any amino acid or has no complementary, tRNA / anticodon or causes the release of the (completed) polypeptide chain ;	1
4(b)	<i>four from</i> Asp and Gln R groups, hydrophilic / AW, and Ile R group, hydrophobic / AW ; <i>ref. to</i> R group interactions and, folding / coiling, of polypeptide chain (to give globular structure) ; <i>ref. to</i> (charged / polar R groups) forming, ionic / hydrogen, bonds ; (amino acids with), hydrophobic / non-polar, R groups, on the inside / AW ; A hydrophobic amino acids (amino acids with), hydrophilic / charged / polar, groups on outside / AW ; A hydrophilic amino acids detail ; e.g. ionic R groups can interact with water, hydrogen bonds form with water <i>ref. to</i> promotes stability / helps to maintain globular structure ; <i>in context of interaction with water / formation of (compact) hydrophobic core</i>	4
4(c)	glycosidic ;	1

Question	Answer	Marks
4(d)	<i>four from</i> 1 both, increase and decrease / peak ; 2 both active, over range of temperature tested / between 20 to 55°C ; 3 free enzyme lower, <u>optimum</u> temperature / temperature for maximum activity ; ora 4 (free) 37°C v (immobilised optimum) 46°C; 5 free enzyme higher, maximum activity / activity at optimum temperature ; ora 6 (free) 8.2 / 8.3, a.u. v 7.5 a.u. ; 7 between 20°C and 39°C the activity of free is higher ; ora 8 between 39°C and 55°C the activity of free is lower ; ora <i>if mp 7 and 8 not gained, allow 1 mark for free activity higher at lower temperatures and lower at higher temperatures (than immobilised) ora</i> 9 comparative data to support mp 7 or 8 ; A comparison at one temperature (stated or use of numerical data) 10 AVP ; e.g. both have same activity at 39°C, similar activity between 20°C and 30°C	4
4(e)	substrate concentration that gives half $V_{\max}$ / AW ; A description of how to use a graph to derive (must refer to substrate concentration)	1
4(f)	<i>two from</i> free / low(er) K_m, has higher affinity for substrate ; free needs a lower concentration of substrate to reach, $V_{\max}$ / maximum activity / $\frac{1}{2} V_{\max}$; takes lower substrate concentration to saturate active sites in free ; free active site more, likely to be saturated with substrate / accessible ; active site (may be) distorted by immobilising ; <i>ref. to</i> substrate passing through matrix when immobilised ; <i>ref. to</i> some product retained within matrix ;	<i>look for ora throughout</i> 2